6 – Divides + QRT

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Cheat sheet

Theorem 4.4.1

Given any integer n and positive integer d, there exist unique integers q and r such that

$$n = dq + r \qquad \text{and} \qquad 0 \leq r < d.$$

Definition

Given an integer n and a positive integer d,

- n **div** $d = q$ (the integer quotient obtained) in the previous theorem,
- n **mod** $d = r$ (the integer remainder obtained) in the previous theorem.

Symbolically, if n is any integer and d is a positive integer, then

$$n \textit{ div } d = q \quad \text{and} \quad n \textit{ mod } d = r$$
$$\iff \qquad n = dq + r \quad \text{and} \quad 0 \leq r < d.$$

Cheat sheet – Divides theorems

① $\forall a, b \in \mathbb{Z}^+$, if $a \mid b$ then $a \leq b$

② Divisors of 1 are 1 & −1

③ $\forall a, b, c \in \mathbb{Z}$, if $a \mid b$ & $b \mid c$, then $a \mid c$

④ If $d \mid a$ and $d \mid b$, then $d \mid (xa + yb)$ $\forall x, y \in \mathbb{Z}$

⑤ $\forall a \in \mathbb{Z}$, $\forall p \in \mathbb{P}$, if $p \mid a$ then $p \nmid (a+1)$

How to show **a | b**?

We just have to show:

b = k*a

Ex. 4.3.16. Prove the statements directly from the definition of divisibility.

For all integers a, b, and c, if $a|b$ and $a|c$ then $a|(b-c)$.

Ex. 4.3.x. Prove the following statements if it is true, and give a counterexample if it is false:

(20) The sum of any three consecutive integers is divisible by 3.
(21) The product of any two even integers is a multiple of 4.
(23) A sufficient condition for an integer to be divisible by 8 is that it be divisible by 16.
(25) For all integers a, b, c, if a divides c then ab divides c.
(30) For all integers a, n, if $a \mid n^2$ and $1 \leq a \leq n$, then $a \mid n$.

Ex. 4.3.26. Prove the statement directly from the definition if it is true, and give a counterexample if it is false.

For all integers a, b, and c, if $ab|c$ then $a|c$ and $b|c$.

Ex. 4.3.39. Suppose that in standard factored form $a = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k}$, where k is a positive integer; $p_1, p_2, \ldots p_k$ are prime numbers; and $e_1, e_2, \ldots, e_k$ are positive integers.

a. What is the standard factored form for a^3?

b. Find the least positive integer k such that $2^4 \cdot 3^5 \cdot 7 \cdot 11^2 \cdot k$ is a perfect cube (i.e., equals an integer to the third power). Write the resulting product as a perfect cube.

Ex. 4.3.48. Prove that for any nonnegative integer n, if the sum of the digits of n is divisible by 3, then n is divisible by 3.

Hint:

3 | 10^j – 1 for all j >= 1 using Geometric Progression

$$\underbrace{\sum_{i=0}^{j} 9 \cdot 10^{i}} = \frac{9\,(10^{j+1} - 1)}{10 - 1} = 10^{j+1} - 1$$

$$= 9 \sum_{i=0}^{j} 10^{i} = 3 \cdot \left(3 \sum_{i=0}^{j} 10^{i} \right)$$

Ex. 4.4.30.

(a) Use the quotient-remainder theorem with $d = 3$ to prove that the product of any two consecutive integers has the form $3k$ or $3k + 2$ for some integer k.

(b) Use the mod notation to rewrite the result of part (a).

Ex. 4.4.31. Prove that for all integers n and m, either $n - m$ and $m + n$ are both even, or they are both odd.

Ex. 4.4.34. Given any integer n, if $n > 3$, could $n, n+2$, and $n+4$ all be primes? Prove or give a counterexample.

SOLUTION . Suppose n is any integer with $n > 3$. By the quotient-remainder theorem with $d = 3$, we know that $n = 3q$, or $n = 3q + 1$, or $n = 3q + 2$ for some integer q. Note that because n is greater than 3; q is greater than 1 or $q = 1$ and $n = 4$ or 5.

Case 1: ($q = 1$ and $n = 4$.)
In this case, n is not prime because $4 = 2 \cdot 2$.

Case 2: ($q = 1$ and $n = 5$.)
In this case, $n + 4 = 9 = 3 \cdot 3$, and so $n + 4$ is not prime.

Case 3: ($q > 1$ and $n = 3q$.)
In this case, n is not prime because it is a product of 3 and q and both 3 and q are greater than 1.

Case 4: ($q > 1$ and $n = 3q + 1$.)
In this case, $n+2 = (3q+1)+2 = 3q+3 = 3(q+1)$. So $n+2$ is not prime because it is a product of 3 and $q+1$ and both 3 and $q+1$ are greater than 1.

Case 5: ($q > 1$ and $n = 3q + 2$.)
In this case, $n+4 = (3q+2)+4 = 3q+6 = 3(q+2)$. So $n+4$ is not prime because it is a product of 3 and $q+2$ and both 3 and $q+2$ are greater than 1.

Hence in all five cases, at least one of n or $n + 2$ or $n + 4$ is not prime. $\square$

Ex. 4.4.35. Prove that the fourth power of any integer has the form $8m$ or $8m+1$ for some integer m.

SOLUTION . Let n be any integer. Then n is either even or odd.

Case 1: $n = 2k$ for some integer k. Then $n^4 = (2k)^4 = 16n^4 = 8(2n^4)$. So, n is of the form $8m$ where $m = 2n^4$.

Case 2: $n = 2l+1$ for some integer l. Then $n^4 = (2l+1)^4 = 16l^4+32l^3+24l^2+8l+1 = 8(2l^4 + 4l^3 + 3l^2 + l) + 1$. So, n is of the form $8m+1$, where $m = 2l^4 + 4l^3 + 3l^2 + l$.

$\square$

Ex. 4.4.50. If m, n, and d are integers, $d > 0$, and $d|(m-n)$, what is the relation between $m \mod d$ and $n \mod d$? Prove your answer.

Bonus Qn: Indirect Proof

Suppose p is prime,

If $p \mid n^2$, then $p \mid n$.

Hint: Contrapositive, then Unique Prime Factorization.